

Drone-Based Human Detection System

for Search & Rescue Operations

Project Requirements Document

Domain	Computer Vision / Aerial Robotics
Objective	Real-time person detection from drone footage across difficult terrains
Core Model	YOLOv8 (Ultralytics)
Terrain Types	Forest / Jungle, Sea / Open Water, Desert
Output	Bounding box overlay with confidence score on live video feed

1. Project Planning & Management

Project Proposal

This project develops a real-time computer vision system that processes live aerial footage captured by drones to automatically detect and localize persons in disaster or emergency scenarios. The system targets three challenging terrain types: forest/jungle, open sea/maritime environments, and desert landscapes.

The detection engine is built on the YOLOv8 architecture, selected for its balance between inference speed and accuracy, which makes it suitable for real-time drone video feeds. Detected persons are highlighted with bounding boxes and confidence scores overlaid directly on the output stream. The primary application is to assist search and rescue (SAR) teams by reducing the human effort needed to scan large or inaccessible areas.

Project Plan — Milestones & Deliverables

- **Phase 1 — Dataset Preparation:** Collect and integrate rescue-focused aerial datasets (SARD, HERIDAL, SeaDronesSee, LADD). Perform annotation conversion to YOLO format, data cleaning, and augmentation for terrain diversity.
- **Phase 2 — Model Training:** Fine-tune YOLOv8 on the combined multi-terrain dataset. Run ablation studies on input resolution, augmentation strategies, and anchor configurations.
- **Phase 3 — Real-Time Pipeline:** Build the inference pipeline accepting RTSP streams or direct video capture. Integrate bounding box and confidence score overlay using OpenCV.
- **Phase 4 — Evaluation:** Benchmark detection accuracy (mAP@0.5) and inference speed (FPS) per terrain type. Analyse failure cases and apply targeted data augmentation.
- **Phase 5 — Delivery:** Finalize documentation, produce a demo video of the live system, and publish model weights and code.

Task Assignment & Roles

- **Computer Vision Engineer(s):** YOLOv8 model training, hyperparameter tuning, and per-terrain accuracy evaluation.
- **Data Engineer(s):** Dataset collection, format conversion, augmentation pipeline, and annotation quality control.
- **Integration Engineer(s):** Real-time inference pipeline, drone feed interface, bounding box overlay, and latency optimization.
- **Documentation Lead:** Technical write-ups, progress tracking, presentation materials, and final report.

Risk Assessment & Mitigation Plan

Risk	Severity	Mitigation
Low accuracy on small/distant persons	High	Use TinyPerson-style augmentation; apply image tiling and multi-scale detection heads.

Dataset imbalance across terrains	High	Augment under-represented terrain classes; apply terrain-weighted loss during training.
Inference latency above real-time threshold	Medium	Optimize with TensorRT/ONNX export; profile bottlenecks early on target hardware.
Drone feed variability (blur, lighting)	Medium	Include random blur, brightness, and noise augmentations in the training pipeline.
Limited real desert/maritime footage	Medium	Supplement with synthetic data generation and domain-adversarial fine-tuning.

KPIs (Key Performance Indicators)

- Detection **mAP@0.5** $\geq$ **0.75** on the combined multi-terrain test set.
- Per-terrain **mAP@0.5** $\geq$ **0.65** on each individual terrain subset (forest, sea, desert).
- Real-time inference speed $\geq$ **15 FPS** on a GPU-equipped machine.
- End-to-end system latency (feed input to annotated output) **< 200 ms**.
- False Negative Rate < 10%** — missing a person in a rescue scenario is the primary failure mode.

2. Literature Review

This section surveys key prior works, purpose-built SAR datasets, and known challenges relevant to aerial person detection across difficult terrains.

Aerial Person Detection with Deep Learning

YOLO-based architectures dominate real-time object detection due to their single-stage pipeline. YOLOv8 (Ultralytics, 2023) introduces anchor-free detection heads and an improved feature pyramid network, achieving state-of-the-art speed–accuracy trade-offs. Studies applying YOLO variants to SAR datasets consistently show that domain-specific fine-tuning on aerial rescue data substantially outperforms direct transfer from ground-level datasets (e.g. COCO), with reported mAP values reaching 93–97% on datasets such as SARD and HERIDAL. This confirms the need for purpose-built rescue training data rather than generic pedestrian datasets.

Rescue-Focused Datasets

- SARD — Search and Rescue Image Dataset (IEEE DataPort, 2021):** Purpose-built for SAR person detection from drone footage. Contains 1,981 manually labelled images covering persons in low/high grass, forest shade, quarries, and roads, with the Corr extension adding fog, snow, ice, and motion-blur variations. Provides YOLO-compatible annotations.
- HERIDAL — Human Detection in Aerial Images:** 1,650 high-resolution (4000×3000 px) UAV images collected over non-urban terrains including mountains, forests, oceans, and deserts. Purpose-built for SAR and widely used as a benchmark; best-reported methods achieve mAP above 93% on this dataset.
- SeaDronesSee (University of Tübingen):** Large-scale maritime UAV dataset for SAR with 8,930 training and 3,750 test images covering object detection, single-object tracking, and multi-object tracking. Directly

addresses the open-water/sea terrain in this project.

- **LADD — Lacmus Drone Dataset (Kaggle, 2023):** Created by real-world SAR volunteer organizations. Focuses on locating missing persons in forested environments from 50–100 m altitude using DJI drones. Contains 1,365 images annotated in both VOC and YOLO format. Unique for capturing authentic SAR operational conditions.
- **POP — Partially Occluded Person Dataset (Nature Scientific Data, 2025):** 8,768 labelled thermal infrared images for partially occluded person detection in wilderness environments. Critical for handling canopy and vegetation occlusion in forest terrains, and relevant for future multimodal extension of this project.

Challenges in Multi-Terrain Detection

Detecting persons across heterogeneous terrains introduces unique per-environment challenges:

- **Forest / Jungle:** Canopy occlusion, camouflage-like foliage backgrounds, and dappled lighting dramatically reduce person visibility. The LADD and HERIDAL datasets document that persons lying on the ground under tree cover are the hardest class to detect.
- **Open Sea / Maritime:** Floating debris, wave patterns, and specular reflections generate false positives. Small head-only visible targets in water require high-resolution inference and low-score threshold tuning.
- **Desert:** Sand-coloured clothing blends with terrain; heat shimmer and overexposure under direct sunlight reduce image contrast. Limited public desert-specific rescue datasets remain a gap in the literature.

Suggested Improvements & Research Directions

- Fusion of RGB and thermal/infrared imaging (as in the POP dataset) for improved detection in low-visibility and occluded conditions.
- Lightweight model distillation (e.g. YOLOv8-nano + TensorRT) for edge deployment on drone-mounted NVIDIA Jetson hardware.
- Integration of multi-object tracking (ByteTrack, DeepSORT) to maintain person identity across frames and reduce redundant alerts.
- Synthetic data generation for underrepresented terrains (desert) via game engines or GAN-based augmentation.
- Active learning pipelines to iteratively improve performance with new operator-labelled footage from field deployments.

Final Grading Criteria

- **Documentation (20%):** Completeness, clarity, and technical depth of all submitted documents.
- **Implementation (40%):** Functionality of the detection system, model accuracy, and real-time performance.
- **Testing & Evaluation (25%):** Rigor of testing across terrain types, use of quantitative metrics, and quality of result analysis.
- **Presentation (15%):** Clarity of oral presentation, quality of demo, and ability to answer technical questions.

3. Requirements Gathering

Stakeholder Analysis

- **Search & Rescue Teams:** Primary end-users. Need reliable, fast detection with minimal false negatives and clear bounding-box output on a monitor or tablet in the field.
- **Drone Operators:** Responsible for flight path. Need the system to run passively without interrupting drone control software.
- **Emergency Management Agencies:** Oversee SAR operations. Need performance metrics and post-mission detection logs for after-action review.
- **Research Supervisors:** Academic or institutional oversight. Need reproducible experiments, documented methodology, and quantitative evaluation.
- **Development Team:** Need clearly defined requirements, quality labelled datasets, and a reproducible training and evaluation pipeline.

User Stories & Use Cases

- As a rescue operator, I want the system to draw bounding boxes around detected persons in the live feed so that I can immediately identify their location without manually scanning footage.
- As a rescue operator, I want confidence scores displayed alongside each detection so that I can prioritize high-confidence sightings during triage.
- As a drone operator, I want inference latency below 200 ms so that the annotated output reflects the near real-time position of the drone.
- As a rescue team lead, I want the system to work across forest, sea, and desert terrains so that a single tool covers all common SAR environments.
- As a research supervisor, I want model accuracy evaluated with standard metrics (mAP, precision, recall, F1) so that performance can be objectively compared against published baselines.

Functional Requirements

- **[FR-01]** The system shall accept a live drone video feed as input (RTSP stream or USB/HDMI capture).
- **[FR-02]** The system shall detect persons in each video frame using a YOLOv8-based model fine-tuned on SAR datasets.
- **[FR-03]** The system shall overlay bounding boxes and confidence scores on detected persons in the output stream using OpenCV.
- **[FR-04]** The system shall support detection across at least three terrain categories: forest/jungle, open sea, and desert.
- **[FR-05]** The system shall allow the model to be retrained or fine-tuned using new terrain-specific datasets via a provided training script.
- **[FR-06]** The system shall log detection events (timestamp, bounding box coordinates, confidence score) to a structured file for post-mission review.

Non-functional Requirements

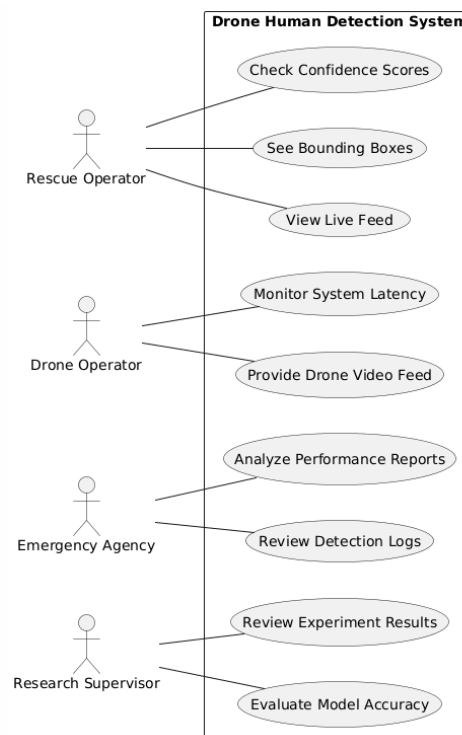
- **Performance:** Inference must run at ≥ 15 FPS with end-to-end latency < 200 ms on an NVIDIA GPU-equipped machine.
- **Accuracy:** The model must achieve $\text{mAP}@0.5 \geq 0.75$ on the combined test set and ≥ 0.65 on each individual terrain subset.
- **Reliability:** The system must operate continuously for the duration of a drone flight (up to 30 minutes) without crashing or performance degradation.
- **Usability:** The annotated video output must be viewable on a standard laptop screen with no software dependency beyond the provided environment.
- **Portability:** The system must run on any machine with Python 3.10+, PyTorch, and a CUDA-enabled NVIDIA GPU.
- **Scalability:** The architecture must allow adding new terrain categories without full retraining, supporting incremental fine-tuning on new terrain data.

4. System Analysis and Design

This section turns ideas into structure. It defines how the system actually works under the hood. The focus is clarity, flow, and execution. Every component has a role. Every step has a reason.

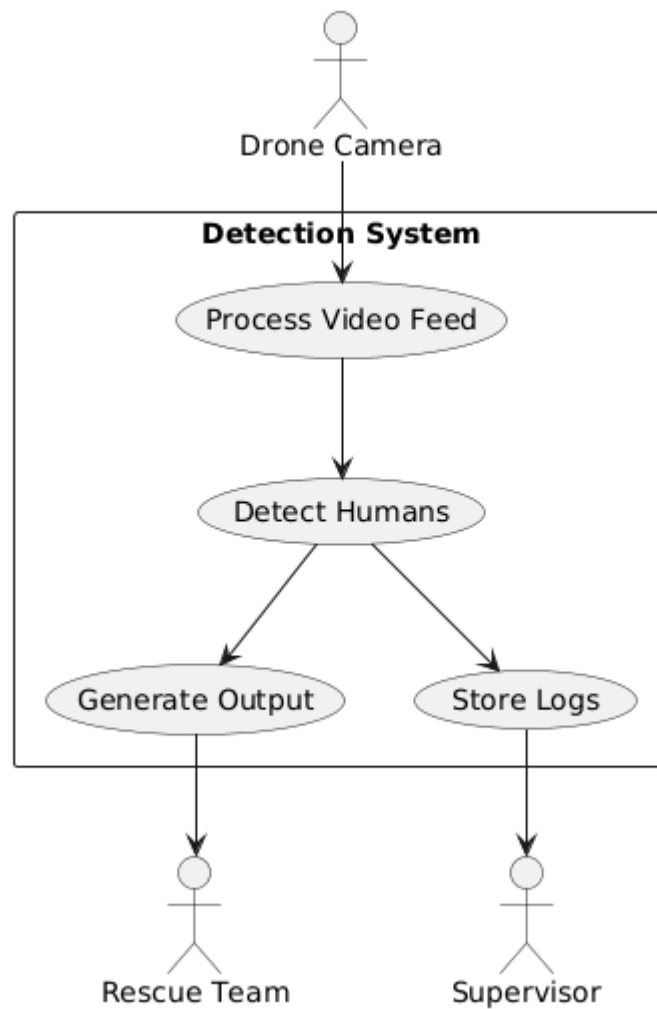
Use-Case Diagram

Shows how users interact with the system and what they expect from it. Focuses on actions like viewing detections and monitoring results.



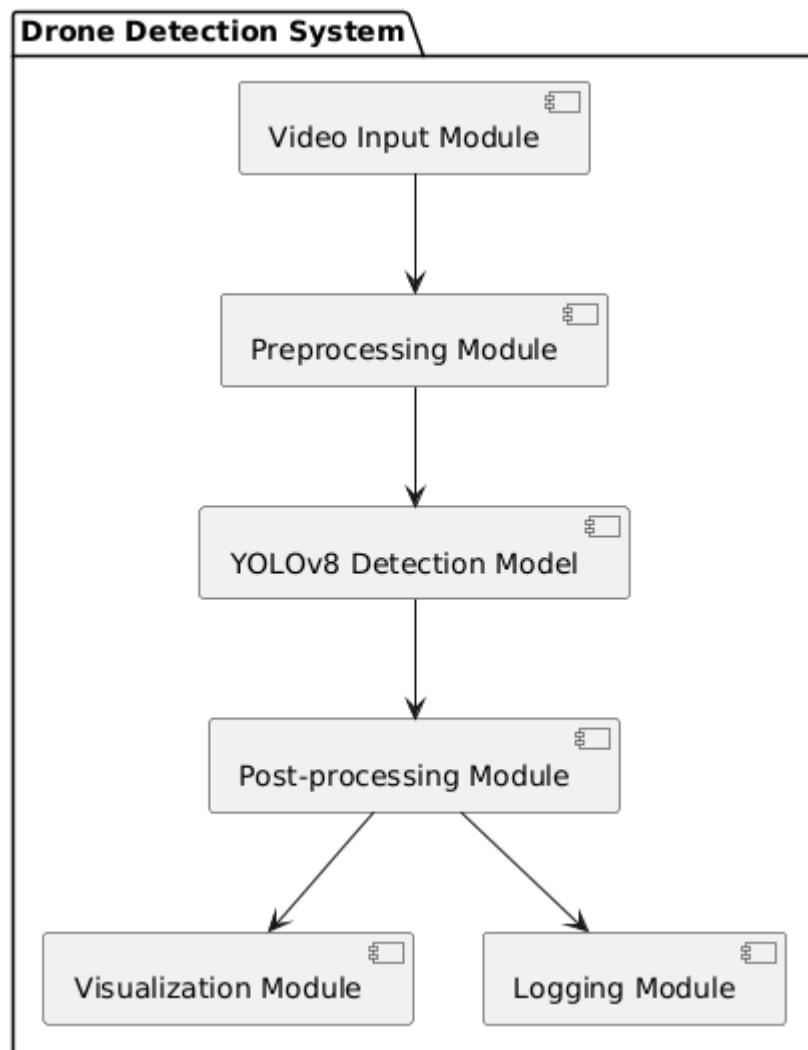
System Context Diagram

Defines system boundaries and external entities. Shows how the drone, users, and system exchange data.



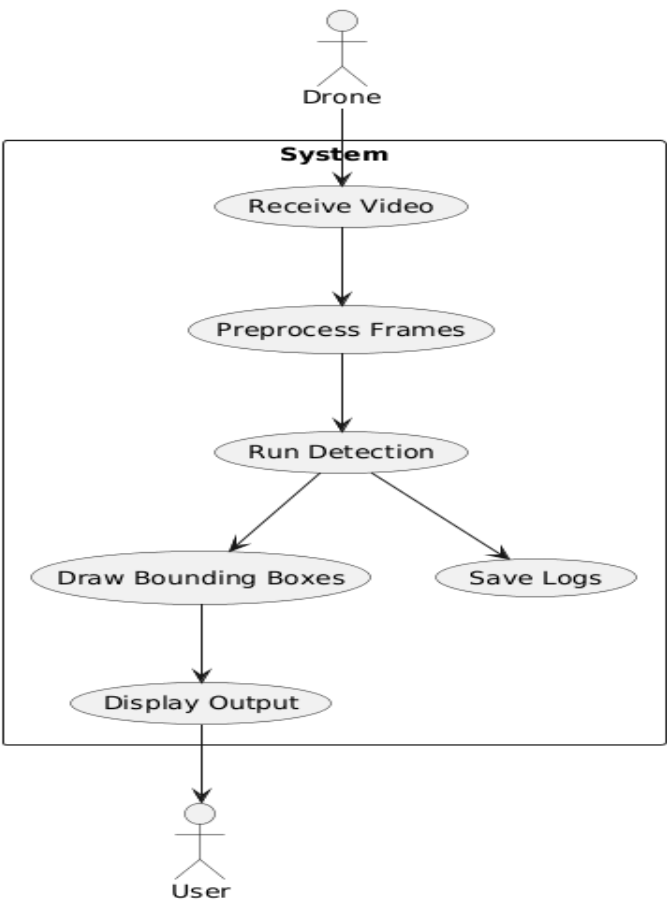
Component Diagram

Breaks the system into core modules and their roles. Highlights how components connect and work together.



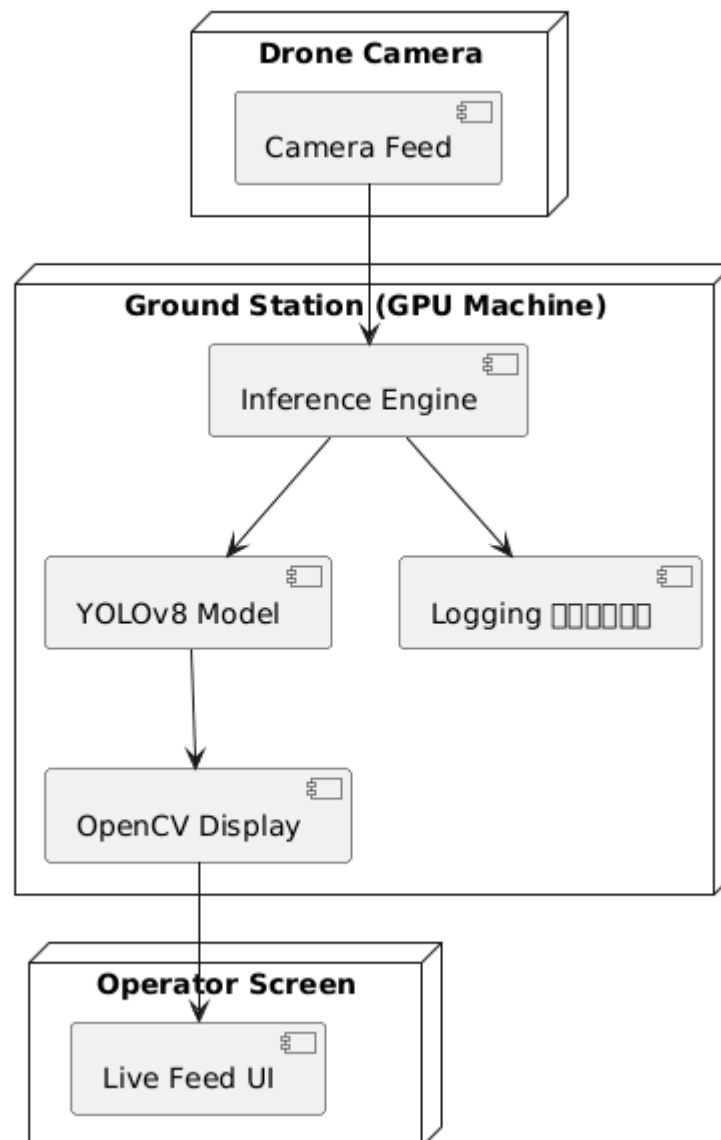
Data Flow Diagram

Tracks how data moves from input to output. Shows processing steps from raw video to final detections.



Deployment Diagram

Illustrates the physical setup of the system. Shows where components run across drone, GPU machine, and user interface.



Model Pipeline Diagram

Shows the full AI workflow from data to deployment. It covers dataset collection, preprocessing, training, evaluation, and final integration into the real time system. This pipeline ensures the model improves continuously and performs reliably across different terrains.

